

LAB 01

Installation of XAMPP

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Course	Database System	Date	20-09226
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1. Objectives

- To understand what XAMPP is and why it is used for web development.
- To download and install XAMPP on a Windows computer.
- To start the Apache and MySQL services using the XAMPP Control Panel.
- To verify the installation using localhost and phpMyAdmin.

2. Introduction

XAMPP is a free and open-source cross-platform web server package. The name stands for Cross-platform (X), Apache (A), MariaDB/MySQL (M), PHP (P) and Perl (P). It bundles everything needed to run dynamic websites and database-driven applications on a local machine, so students can develop and test PHP projects without renting a web server.

Main components: Apache (web server), MariaDB/MySQL (database server), PHP and Perl (scripting languages), and phpMyAdmin (web-based database administration tool).

3. Requirements

Requirement	Details
Operating System	Windows 10 / 11 (64-bit)
Free Disk Space	At least 1 GB
RAM	Minimum 4 GB recommended
Installer	xampp-windows-x64-installer.exe (from apachefriends.org)
Privileges	Administrator access on the PC

4. Procedure

Download the installer from the official website (<https://www.apachefriends.org>), then follow the steps below.

1. **Run the installer.** Double-click the downloaded .exe file. If a User Account Control or antivirus warning appears, click Yes/Allow. On the Setup screen, click Next.

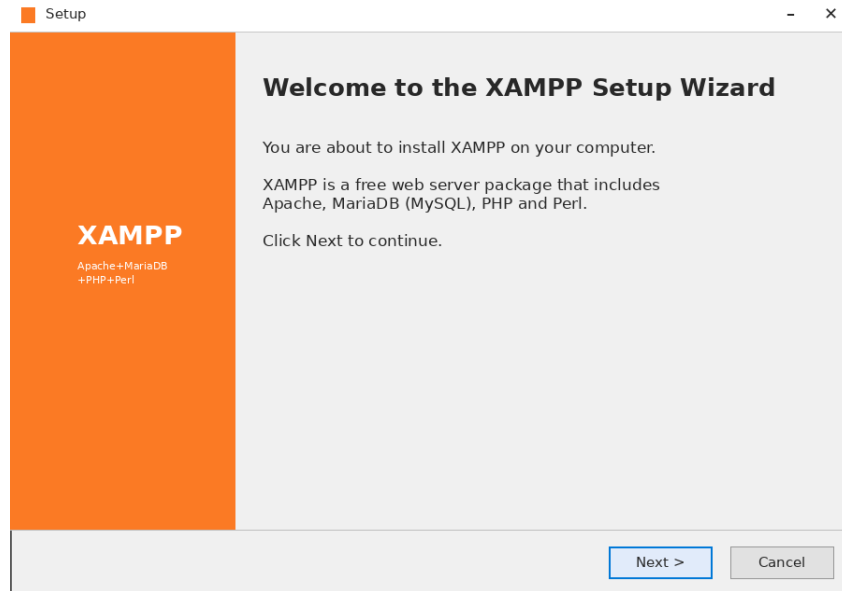


Figure 1: XAMPP Setup Wizard – Welcome screen

2. **Select components.** Keep Apache, MySQL, PHP and phpMyAdmin ticked (the minimum required for this course). Other components are optional. Click Next.

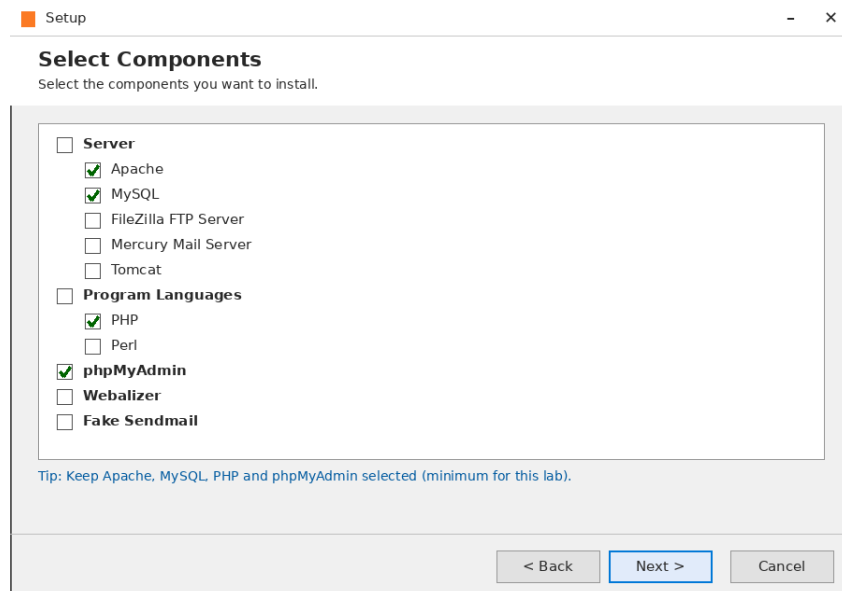


Figure 2: Selecting the components to install

3. **Choose the installation folder.** Use the default location C:\xampp. Do not install into C:\Program Files, because Windows access restrictions can stop XAMPP from working properly. Click Next until the installation begins.

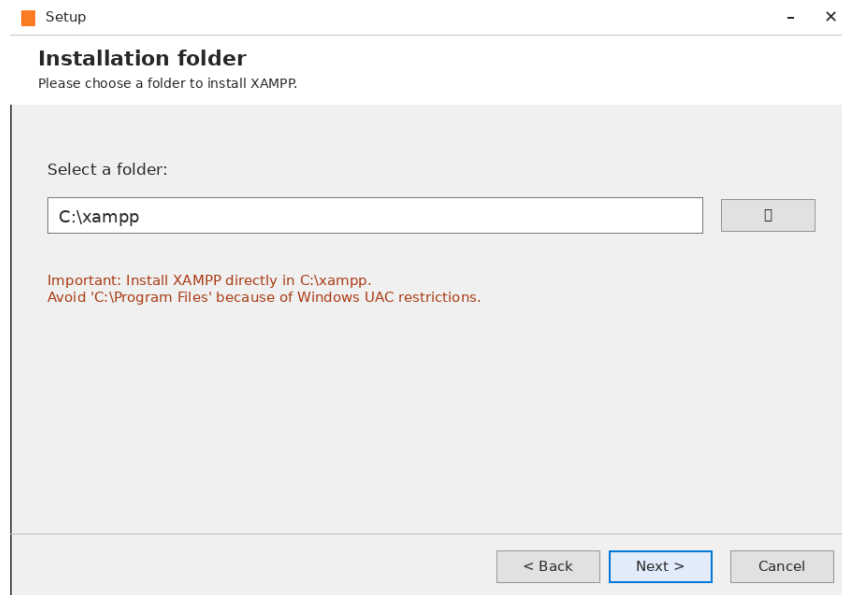


Figure 3: Choosing the installation folder

4. **Wait for the installation to complete.** Files are copied to the chosen folder and a progress bar is shown. This takes a few minutes. When it ends, click Finish and keep “Do you want to start the Control Panel now?” ticked.

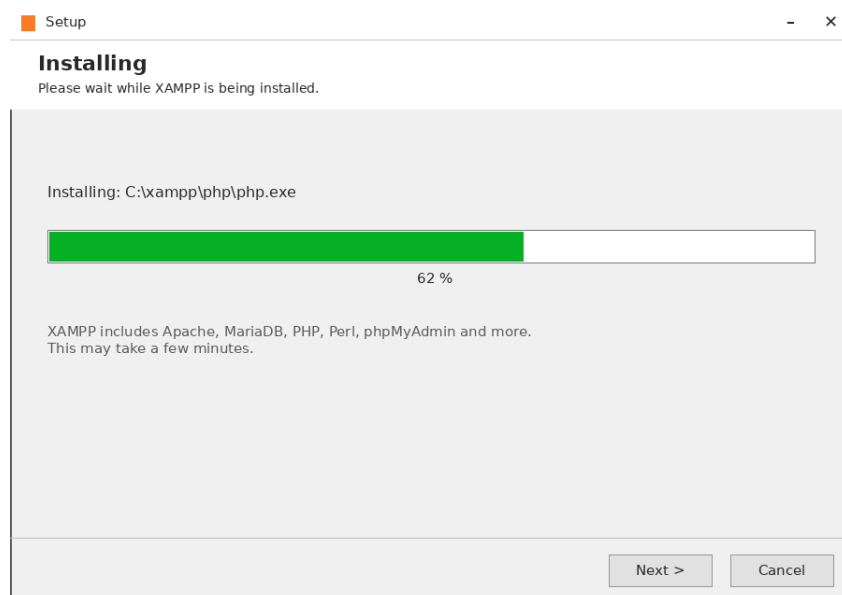


Figure 4: Installation in progress

5. **Start the services.** In the XAMPP Control Panel, click Start next to Apache and MySQL. When they start successfully, their module names turn green and the port numbers (Apache: 80, 443; MySQL: 3306) are displayed.

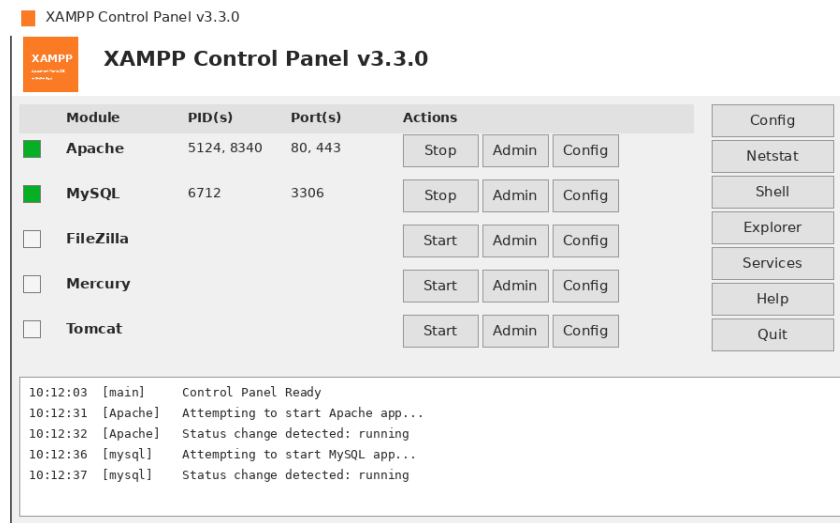


Figure 5: XAMPP Control Panel with Apache and MySQL running

5. Verifying the Installation

1. **Test Apache:** open a web browser and go to <http://localhost>. The XAMPP welcome/dashboard page should appear.
2. **Test MySQL:** go to <http://localhost/phpmyadmin>. The phpMyAdmin interface should open without errors.
3. **Test PHP:** create a file test.php inside C:\xampp\htdocs containing the line `<?php phpinfo(); ?>` and open <http://localhost/test.php>. The PHP information page should be displayed.

6. Common Problems and Solutions

Problem	Solution
Apache does not start (port 80 already in use)	Another program such as Skype, IIS or a second web server is using port 80/443. Close it, or change Apache's port in Config → httpd.conf (e.g. to 8080).
MySQL does not start (port 3306 in use)	Stop any other MySQL service running on the PC, or change the port in my.ini.
Antivirus warning during installation	Temporarily disable the antivirus or allow the installer, then re-run the setup.
Windows Firewall prompt	Allow access for Apache HTTP Server and MySQL on private networks.

7. Conclusion

In this lab, XAMPP was installed and configured on Windows. The Apache and MySQL services were started from the Control Panel and the setup was verified using localhost and phpMyAdmin. This environment will be used in the following labs for developing PHP and database applications.